WORKED SOLUTION

STEP II 2015, Question 9

Mechanics • Rigid Body Rotation

STEP II 2015, Question 9

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/15-s2-q9

The key idea

Everything about this system is controlled by one point: the centre of mass G of the two particles. A rigid body pivoted at A hangs at rest with G vertically below A, and it makes complete revolutions exactly when it has enough energy to carry G over the top. So find where G is and how far it is from A, and both parts follow quickly.

Solution

We model B and C as particles joined by light rigid rods, with a smooth axis at A, so the only forces doing work are the weights.

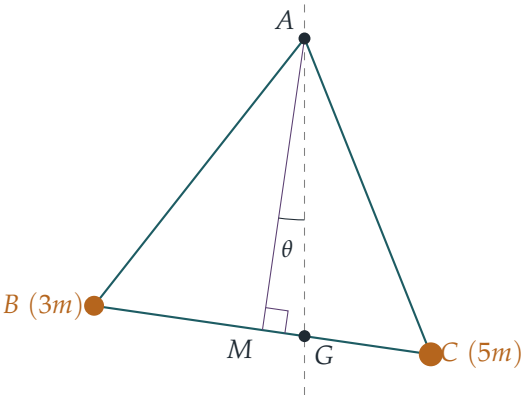
Part (i)

The rods are light, so the centre of mass of the system is the centre of mass of $3m$ at B and $5m$ at C. That point G lies on BC with

$$BG = \frac{5}{8}a, \qquad CG = \frac{3}{8}a.$$

Let M be the midpoint of BC. Then $MG = \frac{5}{8}a - \frac{1}{2}a = \frac{1}{8}a$, on the side of C. Since the triangle is equilateral, $AM \perp BC$ and $AM = \frac{\sqrt{3}}{2}a$, so by Pythagoras in triangle AMG,

$$AG^2 = \frac{3a^2}{4} + \frac{a^2}{64} = \frac{49a^2}{64}, \qquad AG = \frac{7a}{8}.$$



In equilibrium the weight $8mg$ acting at G must have zero moment about A, so G is vertically below A: the line AG is vertical. Now BC is perpendicular to AM, and the horizontal is perpendicular to the vertical AG. Rotating both lines of a pair by a right angle doesn't change the angle between them, so

$$\theta = \angle GAM.$$

In the right-angled triangle AMG (right angle at M),

$$\sin\theta = \frac{MG}{AG} = \frac{a/8}{7a/8} = \frac{1}{7},$$

as required. Note that C, the heavier particle, is the lower end of BC, as it should be.

Part (ii)

Both particles are at distance a from the axis, so they always have the same speed. If the triangle turns with angular speed ω, each particle has speed $a\omega$. Immediately after the impulse, B has speed v, so C does too, and the kinetic energy is

$$\tfrac{1}{2}(3m)v^2 + \tfrac{1}{2}(5m)v^2 = 4mv^2.$$

(Equivalently, the moment of inertia about A is $8ma^2$ and $\omega = v/a$.) The direction of the impulse doesn't matter, since only the speed enters the energy equation below.

? **Ask yourself** Do we need v to be large enough to keep something taut at the top, as for a particle on a string?

No. The rods are rigid and can push as well as pull, so the only requirement is that the angular speed never falls to zero before the system has gone right round. That is a pure energy condition.

The potential energy of the system is $8mg$ times the height of G. Since G moves on a circle of radius $AG = \frac{7}{8}a$ about A, the potential energy is greatest when G is vertically above A, at a height

$$2 \cdot \frac{7a}{8} = \frac{7a}{4}$$

above its starting position. By conservation of energy, at a position where G has risen by h,

$$4m(a\omega)^2 = 4mv^2 - 8mgh.$$

The right-hand side is smallest when $h = \frac{7}{4}a$. If it is positive there, then $\omega \neq 0$ throughout, so ω never changes sign and the system keeps turning the same way, completing revolution after revolution. If it is negative, ω must reach zero before G reaches the top and the system swings back. So the condition is

$$4mv^2 > 8mg \cdot \frac{7a}{4} \iff v^2 > \frac{7ag}{2}.$$

Hence

$$v_0 = \sqrt{\frac{7ag}{2}}.$$

(When $v = v_0$ exactly, G creeps towards the position above A and takes infinitely long to get there, which is why the condition is a strict inequality.)

Checks

- With equal masses G would be at M, giving $\theta = 0$ and $AG = \frac{\sqrt{3}}{2}a$: formulas with $MG = 0$ give exactly that.
- Minimising the potential energy numerically over all orientations gives $\sin\theta = 0.142857\ldots = \frac{1}{7}$.
- Units: ag has units $\mathrm{m}^2\,\mathrm{s}^{-2}$, so v_0 is a speed. For $a = 1\,\mathrm{m}$ and $g = 9.8\,\mathrm{m\,s^{-2}}$, $v_0 \approx 5.86\,\mathrm{m\,s^{-1}}$.
- A step-by-step computer simulation of the motion confirms that the system goes over the top when $v = 1.01v_0$ and swings back when $v = 0.99v_0$.

Common slips

- Taking the rise of G as $2 \cdot \frac{\sqrt{3}}{2}a$ (using AM instead of AG), which gives $v_0^2 = 2\sqrt{3}\,ag$.
- Treating B and C separately and forgetting that the impulse gives C speed v as well.
- Imposing a "string stays taut" condition at the top: rods can be in thrust, so none is needed.

Beyond STEP

The energy argument here is the rigid-body version of the compound pendulum: a body pivoted at a distance d from its centre of mass behaves like a simple pendulum, with the equation of motion $I\ddot{\psi} = -Mgd\sin\psi$. In a first-year course you'll meet the phase plane for this equation, where the critical speed v_0 corresponds to the separatrix

dividing oscillations from complete rotations. Kleppner and Kolenkow, *An Introduction to Mechanics*, treats rigid-body rotation about a fixed axis and the physical pendulum in detail.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.